

TECHNICAL NOTE

Cryptographic Verification as a Mechanism to Satisfy EU AI Act Article 50 Provenance and Marking Requirements

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This note is a technical document. It is not legal advice and it does not determine legal compliance. An APEX PSI seal certifies existence, timestamp and integrity of a record - not the truth of its claims. Organisations must consult qualified legal counsel for their own obligations.

1. THE OBLIGATION

Regulation (EU) 2024/1689 (the EU AI Act), Article 50(2), applicable since 2 August 2026, provides in full:

Providers of AI systems, including general-purpose AI systems, generating synthetic audio, image, video or text content, shall ensure that the outputs of the AI system are marked in a machine-readable format and detectable as artificially generated or manipulated. Providers shall ensure their technical solutions are effective, interoperable, robust and reliable as far as this is technically feasible, taking into account the specificities and limitations of various types of content, the costs of implementation and the generally acknowledged state of the art, as may be reflected in relevant technical standards. This obligation shall not apply to the extent the AI systems perform an assistive function for standard editing or do not substantially alter the input data provided by the deployer or the semantics thereof, or where authorised by law to detect, prevent, investigate or prosecute criminal offences.

Article 50(4) additionally imposes disclosure duties on deployers of deepfake content and of AI-generated text published to inform the public on matters of public interest:

Deployers of an AI system that generates or manipulates image, audio or video content constituting a deep fake, shall disclose that the content has been artificially generated or manipulated. Deployers of an AI system that generates or manipulates text which is published with the purpose of informing the public on matters of public interest shall disclose that the text has been artificially generated or manipulated.

The Act therefore demands three qualities of any technical solution: it must be EFFECTIVE, INTEROPERABLE, and ROBUST AND RELIABLE, as far as technically feasible, having regard to the state of the art.

2. THE GAP

A label or a metadata field stating 'AI-generated' is a claim, not evidence. Labels can be stripped. Metadata can be edited. A screenshot carries neither. A marking scheme whose proof disappears the moment the file is copied cannot, on its own, demonstrate that a provider's technical solution is 'robust and reliable' after the content leaves the provider's environment.

The missing layer is an evidence layer: a machine-readable artifact that (a) travels with or points to the content, (b) cannot be altered without detection, and (c) can be recomputed by any third party without permission, account or fee.

3. THE MECHANISM

APEX PSI supplies that evidence layer, complementary to - not a replacement for - marking schemes such as C2PA or watermarking:

3.1 Canonicalisation. The record (model, input digest, output digest, timestamp, metadata) is serialised per RFC 8785 (JSON Canonicalization Scheme): lexicographic key order, UTF-8, no whitespace ambiguity.

3.2 Seal. seal = SHA-256(canonical record). The seal is deterministic: identical bytes produce an identical seal on any machine.

3.3 Signature. The seal is signed with Ed25519, with an optional post-quantum profile: LMS-W4-SHA256 (NIST SP 800-208) and ML-DSA-65 (FIPS 204).

3.4 Anchoring. Seals are anchored to the Bitcoin chain tip via OpenTimestamps, and included in a public Merkle ledger, giving third-party timestamp evidence without trusting the provider.

3.5 Interoperability. The seal and its verification URL can be embedded as a claim inside C2PA manifests or any metadata container. The mark says 'AI-generated'; the seal proves 'this exact content, at this exact time, unaltered since'.

4. VERIFICATION METHOD

Any party, given the record bytes, recomputes the canonical JSON and its SHA-256. Match = integrity and timestamp confirmed. The same check is served by two independent public doors: ai-governance-standard.com/verify and apex-infrastructure.com/verify. If the doors ever disagree, the correction is public and the finder is credited. Verification is free, requires no account, and is licensed for perpetual public use (MIT).

A marking solution that publishes a recomputable seal per output gives regulators, auditors and courts an objective sufficiency test: the provenance claim survives copying, stripping and time. A solution that publishes only an editable label does not.

5. REFERENCE IMPLEMENTATION AND PRIOR ART

Reference implementation: github.com/kawal393/psi-seal-spec (MIT), zero-dependency Python and TypeScript with byte-identical parity proof and test vectors. Protocol specification: IETF Internet-Draft [draft-singh-psi](#), revision 01 (independent submission). Live public ledger: 5,400+ sealed entries, Bitcoin-anchored.

Worked	example	(Genesis	Zero):	document	SHA-256
f91ba473ee1e88b349d3a4dee18a27d9d8adbc3e1ef1e32aadcd94960a7b7b9b;					receipt
APEX-NTR-C29D90C714C99F96; publicly verifiable at ai-governance-standard.com/genesis .					

6. WHAT THIS NOTE DOES NOT CLAIM

It does not claim that APEX PSI is the only method, nor that use of APEX PSI constitutes compliance. It does not claim that a seal makes content true, lawful or safe. It does not provide legal advice. It states, precisely, that a recomputable, timestamped, independently verifiable seal is a technical mechanism meeting the effectiveness, interoperability and robustness qualities Article 50(2) names - and that the absence of any such recomputable evidence leaves a marking scheme resting on trust alone.

The ledger does not judge. It remembers.